

## TAKUYA KORIYAMA

Website <https://takuyakoriyama.github.io/> Email: tkoriyam@uchicago.edu [Google Scholar](#)

### Education.

- 2024-present: PhD student in Econometrics and Statistics, the University of Chicago, Booth School of Business. Advisor: [Tengyuan Liang](#).
- 2022-2024 PhD student in Statistics, Rutgers University. Advisor: [Pierre C. Bellec](#).
- 2018-2022: BE in Mathematical Engineering, the University of Tokyo.

I'm broadly interested in the mathematical theory and algorithms underlying statistics and machine learning.

### Preprint.

- (1) [A Markov Chain Approach to Preference Alignment](#)  
Takuya Koriyama and Tengyuan Liang  
*arXiv:2606.22652, 2026*
- (2) [Asymptotic Inference for Exchangeable Gibbs Partition](#)  
Takuya Koriyama  
*arXiv:2506.21527, 2025, minor revision at Stochastic Processes and their Applications*
- (3) [Existence of solutions to the nonlinear equations characterizing the precise error of M-estimators](#)  
Pierre C. Bellec and Takuya Koriyama (alphabetical)  
*arXiv:2312.13254, 2023, submitted.*

### Publication.

- (1) [Asymptotics of resampling without replacement in robust and logistic regression](#)  
Pierre C. Bellec and Takuya Koriyama (alphabetical)  
*Bernoulli*, *in press, 2026*
- (2) [Denoising Diffusions with Optimal Transport: Localization, Curvature, and Multi-Scale Complexity](#)  
Tengyuan Liang, Kulunu Dharmakeerthi, and Takuya Koriyama  
*Transactions on Machine Learning Research*, 2835-8856, 2026.
- (3) [Precise asymptotics of bagging regularized M-estimators](#)  
Takuya Koriyama, Pratik Patil, Jin-Hong Du, Kai Tan, and Pierre C. Bellec  
*The Annals of Statistics*, 54(2) 796-833, 2026.
- (4) [Asymptotic mixed normality of maximum likelihood estimator for Ewens–Pitman partition](#)  
Takuya Koriyama, Takeru Matsuda and Fumiyasu Komaki  
*Advances in Applied Probability*, 58(1) 214-234, 2026.
- (5) [Phase transitions for the existence of unregularized M-estimators in single index models](#)  
Takuya Koriyama and Pierre C. Bellec  
*Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*.
- (6) [Error estimation and adaptive tuning for unregularized robust M-estimator](#)  
Pierre C. Bellec and Takuya Koriyama (alphabetical)  
*Journal of Machine Learning Research* 26 (16), 1-40, 2025.
- (7) [Corrected generalized cross-validation for finite ensembles of penalized estimators](#)  
Pierre C. Bellec, Jin-Hong Du, Takuya Koriyama\*, Pratik Patil\* and Kai Tan (alphabetical)  
*Journal of the Royal Statistical Society: Series B*, 87(2), 289-318, 2025.
- (8) [Fully Data-driven Normalized and Exponentiated Kernel Density Estimator with Hyvärinen Score](#)  
Shunsuke Imai, Takuya Koriyama, Shouto Yonekura, Shonosuke Sugasawa and Yoshihiko Nishiyama  
*Journal of Business & Economic Statistics* 43 (1), 110-121, 2025.

**Award.**

- IMS Hannan Graduate Student Travel Award, 2025.
- Travel award, Yale FDS workshop honoring Andrew Barron, 2024.
- Best Ph.D. Qualifying Exam Performance, Department of Statistics, Rutgers University, 2023.
- Student Travel Award, IMS International Conference on Statistics and Data Science, 2022.
- Best Presentation Award, 16th Japan Statistical Society Spring Meeting, 2022.
- Japan Statistical Society Certificate Director's Award, 2022.

**Talk.**

- Proceedings of the 43rd International Conference on Machine Learning (ICML), Seoul, July 2026.
- ISBA satellite Meeting 2026, Tokyo, July 2026.
- Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, July 2025.
- Workshop on Statistical Network Analysis and Beyond (SNAB), Tokyo, June 2025.
- IMS International Conference on Statistics and Data Science, Nice, December 2024.
- Workshop Honoring Andrew Barron: Forty Years at the Interplay of Information Theory, Probability and Statistical Learning, Yale University, April 2024.
- IMS Asia Pacific Rim Meeting, Melbourne, January 2024.
- Seminar talk, Graduate school of Economics, the University of Tokyo, February 2023.
- IMS International Conference on Statistics and Data Science, Florence, December 2022.
- Workshop on random partition of integers, the University of Tokyo, June 2022.
- Japan Statistical Society Meeting, Keio University, March 2022.

**Review.**

- Information and Inference a Journal of the IMA, Journal of the American Statistical Association, Annals of Statistics